

TX4 Blind Portfolio Prediction and Contextual Return

TX4 confirmatory campaign

14 September 2026

Abstract

A frozen IEEE-39 phasor-domain model contains a four-bus SG-to-GFL portfolio that is unstable while all fifteen proper subsets are stable. A reusable all-SG network kernel and four local replacement models predicted this V4 verdict before full-order reveal (16/16, zero false-safe). A contextual return reaches unity at the same controller boundary, and nonlinear phasor-domain TDS preserves the ordering. The result is deliberately scoped: a 512-portfolio V9 benchmark gives 77.15 percent accuracy with 116 false-safe cases, the reduced method is slower, and the reduced boundary run was executed after reveal. The appropriate conclusion is CASE B.

1 Executive verdict

H4={30,33,35,37} is the minimal dynamically unstable portfolio at frozen P4. The numerical record is delivered with scripts, tables, raw traces, hashes, figures, and negative results. No universal or probabilistic generalization is claimed.

2 Scientific question

The test is whether one reusable network kernel plus local SG-to-GFL device substitutions can predict a finite incompatible portfolio and explain it with an exact contextual return. V4 passes; V9 limits the scope.

3 Prior evidence and negative lessons

The frozen TX4 record and prior PowerDynamics/CDW negatives are read-only. This campaign did not rescue those lines or modify the frozen manuscript.

4 Frozen TX4 model

The IEEE-39 phasor-domain DAE uses frozen synchronous-machine, AVR, and ten-state custom GFL semantics. P4 is $(g, k, t, h) = (0.03625, 1.425, 1.5, 1)$. The critical positive-frequency band is 0.3–1.5 Hz. No new controller, governor, limits, EMT interface, or network model was added.

5 Blind-prediction firewall

Before reveal, the code built one all-SG $T_0/K(s)$ and local $\Delta Y_i(s)$ library and enumerated V4 and V9. It did not load FC01, PCV02, FC10, H, kappa, or historical boundary tables. Prediction files were hashed and committed before reveal. The firewall is in `docs/TX4_BLIND_FIREWALL.md`.

6 Baseline kernel and local models

For portfolio S , the audited construction is $K = E^T T_0^{-1} E$, $M_S = D_S K_{SS}$, and $Q_S = (I + D_S K_d)^{-1} D_S K_o$. Local models are equilibrium-matched and reused without post-reveal fitting.

7 Exact reduced closure

The root search used the frozen band, deterministic grids, fixed seeds, root refinement, and a local-factor floor. The factorization is an exact determinant representation under regularity assumptions; it is not an away-from-root approximation to the full spectral abscissa.

8 Blind V4 prediction

The reduced prediction found H4 exactly. Full-order reveal produced 15 stable and one unstable case. Accuracy was 16/16, false-safe 0, false-unstable 0, H exact, and kappa exact at 4. H4 reduced root: $\text{Re}(s) = 0.12700633761 \text{ s}^{-1}$ and 0.62227967095 Hz . Full order: $+0.12700646714 \text{ s}^{-1}$ and 0.62227966958 Hz .

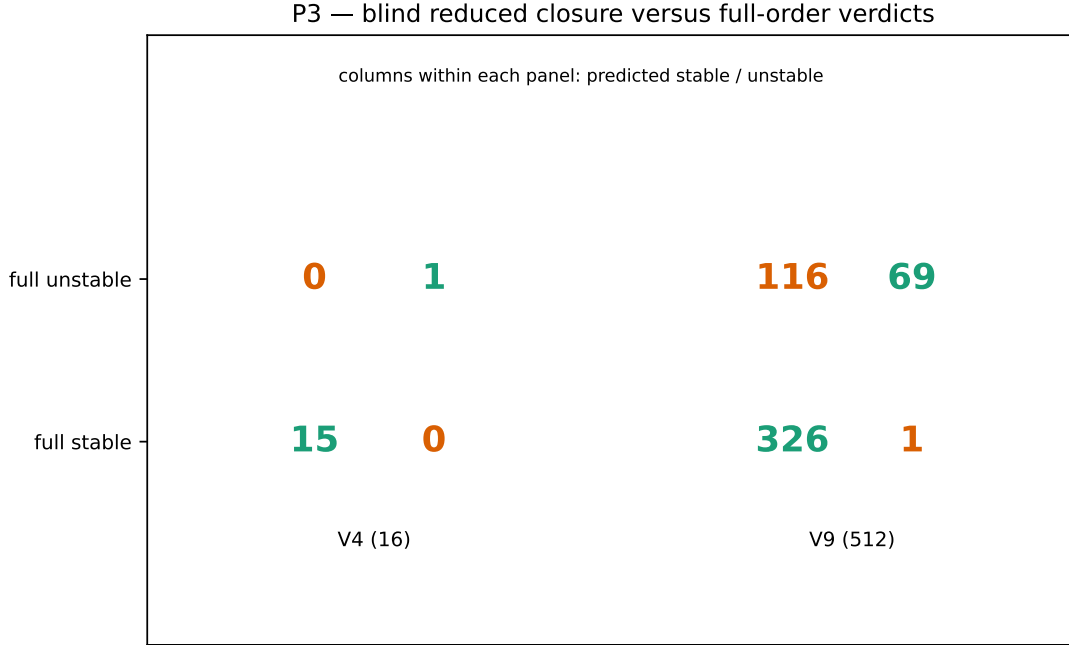


Figure 1: Blind reduced closure versus full-order verdicts.

9 Full-order reveal

V9 truth has 327 stable and 185 unstable cases. The reduced predictor gives 395/512 correct, 116 false-safe, one false-unstable, incorrect H, and exact minimum blocker cardinality 4. The V4 BASE comparison row was restored from the frozen FC01 reference because the PCV02 core table omitted BASE; no prediction was changed.

10 Controller boundary

The reduced-only reproduction gives $g_{\text{return}} = 0.20768138607$ and 0.706424783 Hz . The frozen full-order boundary is $g_{\text{full}} = 0.20768140519$; difference $1.91 \cdot 10^{-8}$. The reduced search ran after reveal, so this is not reported as a strict blind boundary claim.

11 Contextual return

For $R = H \setminus \{i\}$, $G_R = (I + Q_{RR})^{-1}$ and $\mathcal{R}_{i|R} = Q_{iR}G_RQ_{Ri}$. Classical block algebra gives

$$\det(I + Q_H) = \det(I + Q_{RR}) \det(I - \mathcal{R}_{i|R}).$$

The nearest return eigenvalue is $-0.9999999649 - 5.1 \cdot 10^{-9}j$ and maximum Schur residual is $3.44 \cdot 10^{-16}$.

12 Numerical return validation

The nine-case audit with three finite-difference factors produced 27 rows. NumPy and SciPy eigensolvers agreed to displayed precision. Maximum equilibrium algebraic residual was $1.30 \cdot 10^{-12}$. A valid descriptor E, A pair was not exposed and remains NOT AVAILABLE. The return derivative check at $a = g$ has maximum error $7.20 \cdot 10^{-6}$.

13 Minimality separation

retained GFL buses	missing	α	Hz	verdict
30+33+35	37	-0.204688	0.916019	stable
30+33+37	35	-0.175508	0.918500	stable
30+35+37	33	-0.167856	0.914639	stable
33+35+37	30	-0.203009	0.917627	stable
30+33+35+37	–	+0.127006	0.622280	unstable

All fifteen proper subsets pass the stable test at P4.

14 Local versus collective failure

The proper-subset separation is $\eta_H = 0.120831$ in the archived realization; collective $\sigma_{\min} = 2.24 \cdot 10^{-8}$. The reduced boundary run has local $\sigma_{\min} = 0.489$ and local determinant minimum 0.433. This supports collective closure failure, not a local singularity.

15 Fifteen proper subsets

The complete finite certificate is `results/TX4_15_PROPER_SUBSETS.csv`. BASE, singles, pairs, triples, and H4 are retained with closure diagnostics.

16 V9 512-portfolio benchmark

The V9 result is a negative scalability result: 77.1484 percent accuracy, 116 false-safe cases, one false-unstable case, incorrect H recovery, and exact kappa recovery. It prevents the phrase ‘general exact portfolio predictor.’

17 Runtime

set	full seconds	reduced seconds	full/reduced
V4, 16	2.397	4.360	0.550
V9, 512	72.490	309.649	0.234

The reduced method is slower on this hardware.

18 Aggregate MW/MVA counterexample

The fixed rule found six-replacement portfolios with normalized mismatch 0.2513 percent; one is unstable ($\alpha = +324.249$) and one stable ($\alpha = -0.1664$). There is no exact aggregate match under the frozen tolerance. This is supporting evidence, not an exact same-MW/MVA claim.

19 Lower-order screens

Singleton, pair, and triple screens pass while H4 fails. H4 lower-order alpha values are B3 -0.203249, B4 -0.212867, B5 -0.210389, and B5b -0.183673, all predicting stable H4. Exact reduced closure predicts unstable. Third-order alpha and determinant truncation remain separate.

20 Controller remediation

Changing only g from 0.03625 to fixed validated point 0.25 changes H4 from $+0.127006$ to -0.017385 s^{-1} . This is not an economic optimum.

21 Nonlinear TDS

The frozen common disturbance is a +2 percent active-load pulse at bus 20 for 0.2 s. Proper 30+33+35 is stable, H4 at P4 is unstable, and H4 at $g = 0.25$ is stable. All growth signs agree with linear alpha; maximum primary exponent error is $3.51 \cdot 10^{-4} \text{ s}^{-1}$. This is phasor domain TDS, not EMT.

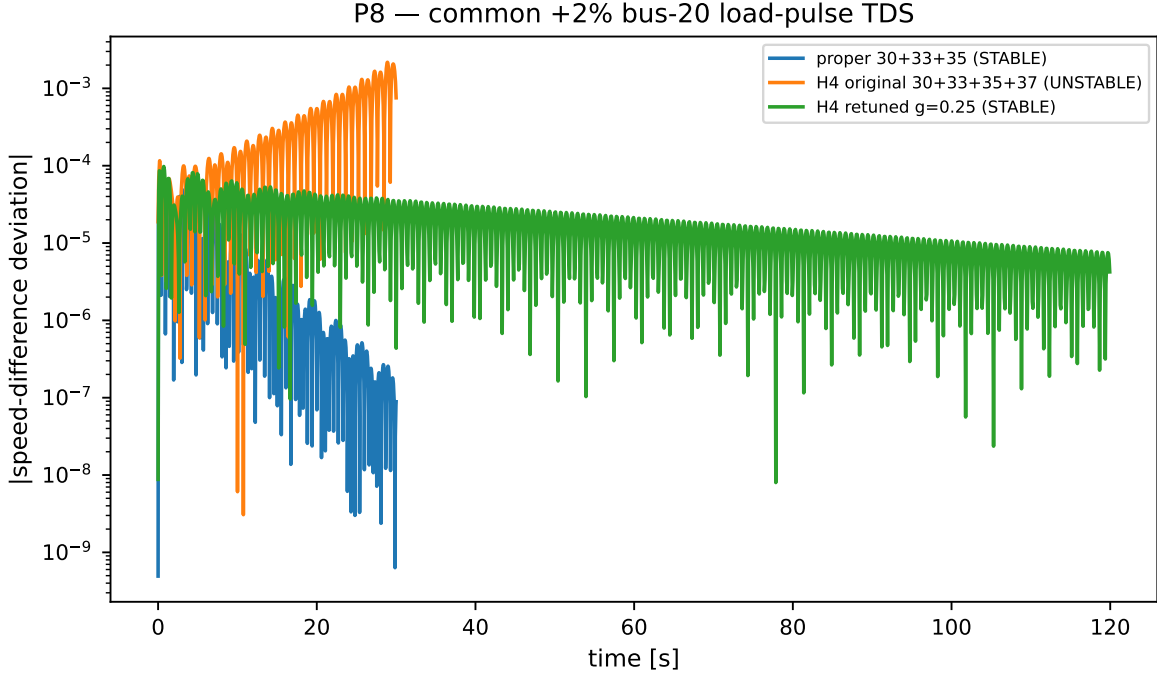


Figure 2: Common-disturbance nonlinear phasor-domain TDS.

22 ANDES

Existing E31 ANDES results were reused without retuning. They provide limited independent phasor support for base and selected portfolio comparisons. Custom-GFL boundary reproduction is NOT TESTED.

23 EMT status

EMT portfolio validation is unresolved and no EMT claim is made.

24 Literature novelty

The focused audit found extensive prior work on grid strength, impedance and return ratios, controller stability regions, and IBR planning. It did not identify in the checked sources the exact combination of finite SG-to-GFL minimality, pre-reveal reduced prediction, and contextual return conditioned on remaining devices. This is a scoped distinction, not a priority claim.

25 Limitations

One IEEE-39 phasor-domain model, one custom GFL implementation, no EMT portfolio validation, limited ANDES equivalence, failed V9 H recovery, slower reduced runtime, and post-reveal timing for the g-boundary search. No universal or probabilistic claim is allowed.

26 Reviewer assessments

Reviewer 1 scores: novelty 7, correctness 8, model adequacy 6, evidence 8, usefulness 7, IAS impact 8, TPWRS readiness 5. Reviewer 2 scores: novelty 7, correctness 8, model adequacy 5, evidence 7, usefulness 7, IAS impact 8, TPWRS readiness 4. Both recommend CASE B poster use with limitations.

27 IAS poster verdict

The recommended title is *SAFE ALONE, UNSAFE TOGETHER*. The poster can show the V4 blind prediction, contextual return, g-only repair, and TDS, but must show V9 and the boundary timing deviation.

28 Transactions path

The package is IAS-ready under a limited predictive claim and not TPWRS-ready. All artifacts are on the branch `research/tx4-blind-portfolio-prediction-final`; no push was made.